

```
#Import necessary libraries
from mlxtend.frequent_patterns import apriori, association_rules
from mlxtend.preprocessing import TransactionEncoder
import pandas as pd

dataset=[
    ['milk', 'bread', 'butter'],
    ['bread', 'butter', 'jam'],
    ['milk', 'bread', 'butter', 'jam'],
    ['butter', 'jam']]

dataset

# Step 1: Preprocessing
te = TransactionEncoder()
te_array = te.fit_transform(dataset)
df = pd.DataFrame(te_array, columns=te.columns_)

# Step 2: Apriori
frequent_itemsets = apriori(df, min_support=0.6, use_colnames=True)

# Step 3: Association rules
num_itemsets = len(frequent_itemsets.itemsets[0])
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.7)

# Output
print("Frequent Itemsets:")
print(frequent_itemsets)

print("\nAssociation Rules:")
print(rules)
```

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated
return datetime.datetime.utcnow().replace(tzinfo=utc)
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Frequent Itemsets:
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	support	itemsets
0	0.75	(bread)
1	1.00	(butter)
2	0.75	(jam)
3	0.75	(butter, bread)
4	0.75	(butter, jam)

Association Rules:

	antecedents	consequents	antecedent support	consequent support	support \
0	(butter)	(bread)	1.00	0.75	0.75
1	(bread)	(butter)	0.75	1.00	0.75
2	(butter)	(jam)	1.00	0.75	0.75
3	(jam)	(butter)	0.75	1.00	0.75

	confidence	lift	representativity	leverage	conviction	zhangs_metric \
0	0.75	1.0	1.0	0.0	1.0	0.0
1	1.00	1.0	1.0	0.0	inf	0.0
2	0.75	1.0	1.0	0.0	1.0	0.0
3	1.00	1.0	1.0	0.0	inf	0.0

	jaccard	certainty	kulczynski
0	0.75	0.0	0.875
1	0.75	0.0	0.875
2	0.75	0.0	0.875
3	0.75	0.0	0.875

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/usr/local/lib/python3.12/dist-packages/mlxtend/frequent_patterns/association_rules.py:186: RuntimeWarning: invalid value encountered in divide
cert_metric = np.where(certainty_denom == 0, 0, certainty_num / certainty_denom)
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